

AmazonKart Black Friday Infrastructure

Step-by-Step Implementation Runbook with Console Steps, Linux Commands, Validation and DR

Item	Value
Region	us-east-1 (N. Virginia)
VPC	AmazonKart-VPC - 10.0.0.0/16
Production EC2	AmazonKart-Prod-Web01 - i-01d773eccc68d0c5f
Production OS	Amazon Linux 2023
Static S3	amazonkart-static-2026-28-08
Backup S3	amazonkart-backup-2026-28-08
Golden AMI	AmazonKart-Prod-Golden-AMI - ami-05f02a718c6d21b63
Tested DR RTO	Approximately 3 minutes 9 seconds - PASS (<15 min)

Purpose: this document records, in implementation order, what was configured during the AmazonKart project and the commands used on the EC2 host. AWS Console operations are explicitly marked as Console Steps; shell commands are shown in code blocks.

1. Project Scope and Service Restriction

The Black Friday challenge required a secure, reliable and cost-conscious AWS environment. We followed the strict allowed-services list: EC2, VPC, S3, IAM and CloudWatch.

- Used: Amazon EC2, Amazon VPC, Amazon S3, AWS IAM, Amazon CloudWatch.
- Not used: ALB, Auto Scaling, RDS, Route 53, NAT Gateway, AWS Backup, Systems Manager, CloudFormation, Terraform.
- The challenge also requested SNS email notification, but SNS was not in the allowed-service list. Therefore CloudWatch alarms were created without SNS actions and the contradiction is documented.
- A single t3.micro cannot realistically guarantee 100,000+ concurrent-user capacity; this is a challenge-constrained design, not a production-scale HA architecture.

2. Naming Plan

Resource	Name / ID
VPC	AmazonKart-VPC / vpc-02b5202bbc5f9861e
Public Subnet	AmazonKart-Public-Subnet-1 / subnet-0a254617a4c3d9197
Private Subnet	AmazonKart-Private-Subnet-1 / subnet-0607efd4f4f728ed4
Internet Gateway	igw-05de8d6969ecff270
Public NACL	AmazonKart-Public-NACL / acl-0e4b7edca14577380
Private NACL	AmazonKart-Private-NACL / acl-042bd5c057b67bf13
Web Security Group	AmazonKart-Web-SG / sg-0afcf9ae96a2fd000
EC2	AmazonKart-Prod-Web01 / i-01d773eccc68d0c5f
IAM EC2 Role	AmazonKart-EC2-Production-Role
CloudWatch Dashboard	AmazonKart-Production-Dashboard

3. Create the Custom VPC

3.1 Console Steps

1. Open AWS Console -> VPC -> Your VPCs -> Create VPC.
2. Select VPC only.
3. Name: AmazonKart-VPC.
4. IPv4 CIDR: 10.0.0.0/16.
5. Tenancy: Default.
6. Create VPC.

3.2 Validation

Final VPC: vpc-02b5202bbc5f9861e with CIDR 10.0.0.0/16.

4. Create Public and Private Subnets

4.1 Public Subnet

7. VPC -> Subnets -> Create subnet.
8. VPC: AmazonKart-VPC.
9. Name: AmazonKart-Public-Subnet-1.
10. Availability Zone: us-east-1a.
11. CIDR: 10.0.1.0/24.
12. After creation, enable Auto-assign public IPv4 for this public subnet where required.

4.2 Private Subnet

13. Create another subnet in AmazonKart-VPC.
14. Name: AmazonKart-Private-Subnet-1.
15. Availability Zone: us-east-1a.
16. CIDR: 10.0.2.0/24.
17. Do not enable automatic public IPv4 assignment for the private subnet.

4.3 Final Values

Subnet	CIDR	Purpose
AmazonKart-Public-Subnet-1	10.0.1.0/24	Production internet-facing EC2
AmazonKart-Private-Subnet-1	10.0.2.0/24	Isolated private tier / future private workload

5. Create and Attach the Internet Gateway

5.1 Console Steps

18. VPC -> Internet Gateways -> Create internet gateway.
19. Create the IGW and attach it to AmazonKart-VPC.

Internet Gateway ID: igw-05de8d6969ecff270

6. Configure the Public Route Table

6.1 Console Steps

20. VPC -> Route Tables -> create/select the public route table.
21. Associate AmazonKart-Public-Subnet-1 with this route table.
22. Edit routes and add the default internet route.

Destination	Target
10.0.0.0/16	local
0.0.0.0/0	igw-05de8d6969ecff270

The private subnet was kept isolated; no NAT Gateway was used because it was not in the allowed service set.

7. Configure Network ACLs

7.1 Public NACL

NACL: acl-0e4b7edca14577380 / AmazonKart-Public-NACL

Inbound rules used:

Rule	Protocol / Port	Source	Action
100	TCP 22	223.228.29.178/32	ALLOW
110	TCP 80	0.0.0.0/0	ALLOW
120	TCP 443	0.0.0.0/0	ALLOW
130	TCP 1024-65535	0.0.0.0/0	ALLOW
*	All	0.0.0.0/0	DENY

Outbound rules used:

Rule	Protocol / Port	Destination	Action
100	TCP 80	0.0.0.0/0	ALLOW
110	TCP 443	0.0.0.0/0	ALLOW
210	TCP 1024-65535	0.0.0.0/0	ALLOW
*	All	0.0.0.0/0	DENY

The ephemeral port rule was necessary because Network ACLs are stateless: return traffic for outbound HTTPS/HTTP sessions must be explicitly allowed.

7.2 Private NACL

NACL: acl-042bd5c057b67bf13 / AmazonKart-Private-NACL

We separated the private subnet from the Public NACL and associated it with a dedicated restrictive NACL.

Direction	Rule	Protocol / Port	Peer	Action
Inbound	100	TCP 1024-65535	10.0.1.0/24	ALLOW
Inbound	*	All	All	DENY
Outbound	100	TCP 1024-65535	10.0.1.0/24	ALLOW
Outbound	*	All	All	DENY

Association verified: AmazonKart-Private-Subnet-1 / subnet-0607efd4f728ed4 uses AmazonKart-Private-NACL.

8. Create the Web Security Group

8.1 Console Steps

23. EC2 -> Security Groups -> Create security group.
24. Name: AmazonKart-Web-SG.
25. VPC: AmazonKart-VPC.
26. Configure only the required inbound ports.

Type	Port	Source
SSH	22	223.228.29.178/32
HTTP	80	0.0.0.0/0

HTTPS	443	0.0.0.0/0
-------	-----	-----------

An older SSH /32 rule was removed after the administrator public IP changed. No SSH rule from 0.0.0.0/0 remained.

9. IAM: Teams, Users, MFA and EC2 Role

9.1 IAM Groups

Created the following functional groups:

Group	Purpose
AmazonKart-Operations	EC2 operations, CloudWatch visibility, backup access
AmazonKart-Security	Security/IAM/VPC/EC2/S3/CloudWatch visibility
AmazonKart-Audit	Read-only audit visibility
AmazonKart-Intern	Limited EC2 and CloudWatch read access

9.2 IAM Users

User	Group
AmazonKart-Ops-User	AmazonKart-Operations
AmazonKart-Sec-User	AmazonKart-Security
AmazonKart-Audit-User	AmazonKart-Audit
AmazonKart-Intern-User	AmazonKart-Intern

27. Create each IAM user in AWS Console.

28. Enable console access using an initial generated password and require password change.

29. Add each user to the correct group.

30. Configure MFA for all four users using Passkey/Security Key.

AdministratorAccess and PowerUserAccess were not granted to everyone.

9.3 EC2 Production Role

31. IAM -> Roles -> Create role -> Trusted entity: AWS service -> EC2.

32. Role name: AmazonKart-EC2-Production-Role.

33. Attach the S3 permissions required by the production server and backup/restore process.

The captured S3 portion of the EC2 role policy was:

```
{
  "Version": "2012-10-17",
  "Statement": [
    {
      "Sid": "ReadStaticContent",
      "Effect": "Allow",
      "Action": ["s3:ListBucket"],
      "Resource": "arn:aws:s3::amazonkart-static-2026-28-08"
    },
    {
      "Sid": "GetStaticObjects",
      "Effect": "Allow",
      "Action": ["s3:GetObject"],
      "Resource": "arn:aws:s3::amazonkart-static-2026-28-08/*"
    }
  ]
}
```

```

    "Sid": "ListBackupBucket",
    "Effect": "Allow",
    "Action": ["s3:ListBucket"],
    "Resource": "arn:aws:s3::amazonkart-backup-2026-28-08"
  },
  {
    "Sid": "UploadBackups",
    "Effect": "Allow",
    "Action": ["s3:PutObject", "s3:GetObject"],
    "Resource": "arn:aws:s3::amazonkart-backup-2026-28-08/daily/*"
  }
]
}

```

Important: s3:GetObject on the backup daily/* prefix was added during the DR test after a 403 restore failure. This became part of the final least-privilege recovery policy.

CloudWatch Agent also required permissions to publish custom metrics and log events. The exact CloudWatch policy JSON used was not fully captured in the recorded console evidence, so this document does not falsely reproduce an exact policy body.

10. Launch the Production EC2 Instance

10.1 Console Configuration

Setting	Value
Name	AmazonKart-Prod-Web01
AMI	Amazon Linux 2023 / ami-0332d564d76dbd8d6
Instance type	t3.micro
Key pair	AmazonKart-Prod-Key
VPC	AmazonKart-VPC
Subnet	AmazonKart-Public-Subnet-1
Private IP	10.0.1.209
Security Group	AmazonKart-Web-SG
IAM Role	AmazonKart-EC2-Production-Role
Metadata	IMDSv2 Required
Detailed monitoring	Disabled - default EC2 CPU metric at 5-minute resolution

34. Launch the instance in the public subnet with a public IPv4 address.

35. Wait until both EC2 status checks pass.

11. Connect to EC2 and Verify the Host

11.1 SSH from Windows

PEM location used:

```
C:\Users\mbaru\Downloads\AmazonKart-Prod-Key.pem
```

Generic SSH command used whenever the current public IP was known:

```
ssh -i "C:\Users\mbaru\Downloads\AmazonKart-Prod-Key.pem" ec2-user@<CURRENT_PUBLIC_IP>
```

Because no Elastic IP was used, the public IP can change after stop/start. When it changes, update the SSH /32 rule in both the Security Group and Public NACL.

11.2 Initial Verification Commands

```
whoami
hostname
cat /etc/os-release
ip addr
df -h
```

12. Harden SSH

We verified effective sshd configuration rather than relying only on the config file.

```
sudo sshd -T | grep -E 'passwordauthentication|pubkeyauthentication'
```

Verified output:

```
pubkeyauthentication yes
passwordauthentication no
```

Result: SSH key authentication enabled and password authentication disabled.

13. Install and Configure Apache

13.1 Install Apache

```
sudo dnf install -y httpd
sudo systemctl enable --now httpd
sudo systemctl status httpd
```

13.2 Create the Website

The website was stored under /var/www/html. A clean reconstruction of the final page is:

```
sudo tee /var/www/html/index.html > /dev/null <<'EOF'
<!DOCTYPE html>
<html>
<head>
  <meta charset="UTF-8">
  <title>AmazonKart</title>
</head>
<body>
  <h1>Welcome to AmazonKart</h1>
  <h3>Black Friday Production Website</h3>
  <p>Server: AmazonKart-Prod-Web01</p>
  <h4>Static Content from Private S3</h4>
  
</body>
</html>
EOF
```

13.3 Verify Apache

```
curl -I http://localhost
sudo systemctl is-enabled httpd
sudo systemctl is-active httpd
```

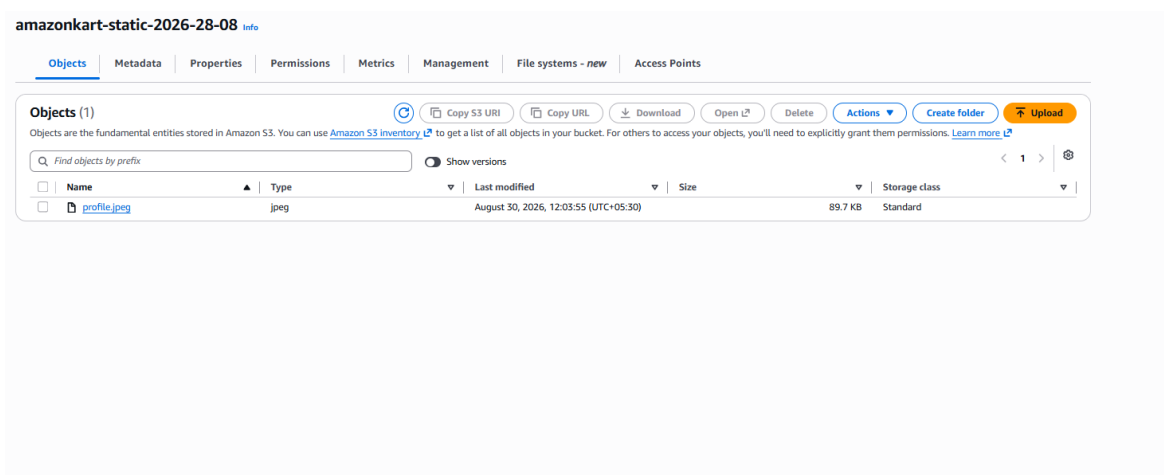
Expected: HTTP/1.1 200 OK and httpd active/enabled.

14. Create and Secure S3 Buckets

14.1 Static Content Bucket

Bucket: amazonkart-static-2026-28-08

- 36. S3 -> Create bucket in us-east-1.
- 37. Enable Bucket Versioning.
- 38. Default encryption: SSE-S3.
- 39. Block all public access: ON.
- 40. Static website hosting: Disabled.
- 41. Upload profile.jpeg.



Evidence: profile.jpeg uploaded to the static S3 bucket.

14.2 Backup Bucket

Bucket: amazonkart-backup-2026-28-08

The production role was permitted to upload backups into the daily/ prefix, and later to read that prefix for DR restore.

15. Retrieve Private S3 Static Content from EC2

15.1 Create Local Static Directory

```
sudo mkdir -p /var/www/html/static
```

15.2 Copy Object from Private S3

```
sudo aws s3 cp s3://amazonkart-static-2026-28-08/profile.jpeg /var/www/html/static/profile.jpeg
```

15.3 Verify Object and HTTP Delivery

```
ls -lh /var/www/html/static/profile.jpeg
curl -I http://localhost/static/profile.jpeg
```

Observed object size was approximately 90 KB and Apache returned HTTP 200.

Welcome to AmazonKart

Black Friday Production Website

Server: AmazonKart-Prod-Web01

Static Content from Private S3



Evidence: production website renders the image retrieved from private S3.

16. Test S3 Least-Privilege Behavior

We verified that the EC2 role could perform intended actions but could not write into the static bucket.

```
aws s3 ls s3://amazonkart-static-2026-28-08
echo "test" > /tmp/test.txt
aws s3 cp /tmp/test.txt s3://amazonkart-backup-2026-28-08/daily/test.txt
aws s3 cp /tmp/test.txt s3://amazonkart-static-2026-28-08/test.txt
```

Expected final command: AccessDenied. This was useful least-privilege evidence because the production role was read-only for static objects.

17. Install the CloudWatch Agent

17.1 Install

```
sudo dnf install -y amazon-cloudwatch-agent
```

17.2 Create Metrics Configuration

The project used custom namespace AmazonKart/Production for memory and disk metrics. The following reflects the final metric intent:

```
sudo tee /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json > /dev/null <<'EOF'
{
  "agent": {
    "metrics_collection_interval": 60,
    "run_as_user": "root"
  },
  "metrics": {
    "namespace": "AmazonKart/Production",
    "append_dimensions": {
      "InstanceId": "${aws:InstanceId}"
    },
    "metrics_collected": {
      "mem": {
        "measurement": [
          "mem_used_percent"
        ],
        "metrics_collection_interval": 60
      },
      "disk": {
        "measurement": [
          "used_percent"
        ],
        "metrics_collection_interval": 60,
        "resources": [
          "/"
        ]
      }
    }
  }
}
EOF
```

17.3 Load the Configuration

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl \
-a fetch-config \
-m ec2 \
-s \
-c file:/opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.json
```

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a status
```

The active fetched configuration was later found under the agent's amazon-cloudwatch-agent.d directory. We backed it up before extending it for logs.

18. Enable System Log Files on Amazon Linux 2023

Amazon Linux 2023 primarily uses journald. To collect /var/log/messages and /var/log/secure in the challenge's requested file style, rsyslog was installed.

```
sudo systemctl status rsyslog
sudo dnf install -y rsyslog
sudo systemctl enable --now rsyslog
sudo systemctl status rsyslog
```

18.1 Create a Test System Log Entry

```
sudo logger "AmazonKart CloudWatch system log test"
sudo tail -n 20 /var/log/messages
```

19. Add Apache, Auth and System Logs to CloudWatch Agent

19.1 Verify Source Files

```
sudo ls -lh /var/log/httpd/
sudo ls -lh /var/log/secure /var/log/messages
```

Amazon Linux 2023 file mapping used:

Challenge Name	Amazon Linux 2023 Actual File
access.log	/var/log/httpd/access_log
error.log	/var/log/httpd/error_log
auth.log	/var/log/secure
messages	/var/log/messages

19.2 Backup the Current Agent Config

```
sudo cp /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.d/file_amazon-cloudwatch-agent.json \
/opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.d/file_amazon-cloudwatch-agent.json.backup
```

19.3 Final Metrics + Logs Configuration

A practical reconstruction of the working final configuration is shown below. If rebuilding, validate the JSON before restarting the agent.

```
{
  "agent": {
    "metrics_collection_interval": 60,
    "run_as_user": "root"
  },
  "metrics": {
    "namespace": "AmazonKart/Production",
    "append_dimensions": {
      "InstanceId": "${aws:InstanceId}"
    },
    "metrics_collected": {
      "mem": {
        "measurement": ["mem_used_percent"],
        "metrics_collection_interval": 60
      },
      "disk": {
        "measurement": ["used_percent"],
        "metrics_collection_interval": 60,
        "resources": ["/"]
      }
    }
  },
  "logs": {
    "logs_collected": {
      "files": {
        "collect_list": [

```

```

    "file_path": "/var/log/httpd/access_log",
    "log_group_name": "/amazonkart/production/apache/access",
    "log_stream_name": "{instance_id}"
  },
  {
    "file_path": "/var/log/httpd/error_log",
    "log_group_name": "/amazonkart/production/apache/error",
    "log_stream_name": "{instance_id}"
  },
  {
    "file_path": "/var/log/secure",
    "log_group_name": "/amazonkart/production/auth",
    "log_stream_name": "{instance_id}"
  },
  {
    "file_path": "/var/log/messages",
    "log_group_name": "/amazonkart/production/system",
    "log_stream_name": "{instance_id}"
  }
]
}
}
}
}
}

```

19.4 Validate and Restart

```
python3 -m json.tool /opt/aws/amazon-cloudwatch-agent/etc/amazon-cloudwatch-agent.d/file_amazon-cloudwatch-agent.json > /dev/null && echo "JSON VALID"
```

```
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a status
```

```
sudo systemctl restart amazon-cloudwatch-agent
```

```
sudo systemctl status amazon-cloudwatch-agent
```

All four CloudWatch log groups later showed real events. Retention was set to 1 month (30 days).

20. Configure CloudWatch Metrics, Alarms and Dashboard

20.1 CPU Metric

42. CloudWatch -> Metrics -> EC2 -> Per-Instance Metrics.

43. Select CPUUtilization for i-01d773eccc68d0c5f.

Because EC2 detailed monitoring was disabled, CPU used the standard 5-minute period.

20.2 Create CPU Alarm

Setting	Value
Alarm	AmazonKart-CPU-High-75
Metric	EC2 CPUUtilization
Statistic	Average
Period	5 minutes
Threshold	> 75%
Datapoints	1 of 1
Missing data	Missing
Notification	No SNS action because SNS was prohibited by the challenge's strict allow-list

20.3 Create Memory Alarm

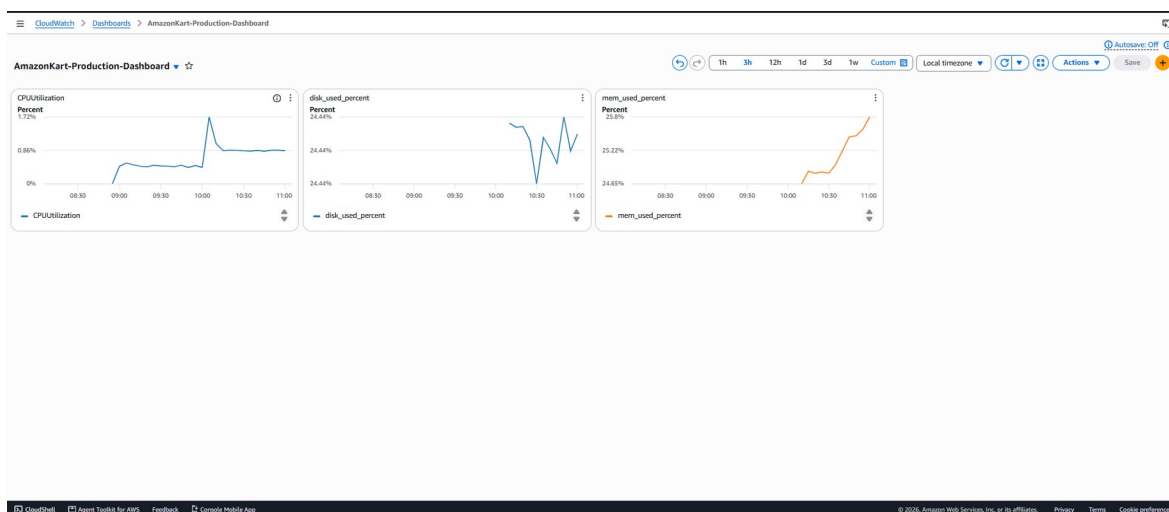
Setting	Value
Alarm	AmazonKart-Memory-High-80
Metric	AmazonKart/Production - mem_used_percent
Statistic	Average
Period	1 minute
Threshold	> 80%
Datapoints	1 of 1

20.4 Create Disk Alarm

Setting	Value
Alarm	AmazonKart-Disk-High-85
Metric	AmazonKart/Production - disk_used_percent
Filesystem	/
Device	nvme0n1p1
fstype	xfs
Statistic	Average
Period	1 minute
Threshold	> 85%
Datapoints	1 of 1

20.5 Dashboard

44. CloudWatch -> Dashboards -> Create dashboard.
45. Name: AmazonKart-Production-Dashboard.
46. Add a line widget for CPUUtilization.
47. Add a line widget for mem_used_percent.
48. Add a line widget for root disk disk_used_percent.



Evidence: AmazonKart production dashboard with CPU, memory and disk metrics.

21. Configure CloudWatch Log Retention

49. CloudWatch -> Logs -> Log groups.

50. For each AmazonKart log group, select Actions -> Retention settings.

51. Set retention to 1 month / 30 days.

Log Group	Retention
/amazonkart/production/apache/access	30 days
/amazonkart/production/apache/error	30 days
/amazonkart/production/auth	30 days
/amazonkart/production/system	30 days

22. Configure the Daily Website Backup

22.1 Set Server Timezone to IST

```
timedatectl
sudo timedatectl set-timezone Asia/Kolkata
timedatectl
```

This made the cron expression 0 0 * * * correspond directly to 12:00 AM IST.

22.2 Create the Backup Script

```
sudo tee /usr/local/bin/amazonkart-backup.sh > /dev/null <<'EOF'
#!/bin/bash

BACKUP_DIR="/tmp"
SOURCE="/var/www/html"
BUCKET="s3://amazonkart-backup-2026-28-08/daily"
DATE=$(date +"%Y-%m-%d_%H-%M-%S")
FILE="$BACKUP_DIR/amazonkart-web-$DATE.tar.gz"
LOG="/var/log/amazonkart-backup.log"

tar -czf "$FILE" "$SOURCE"

if aws s3 cp "$FILE" "$BUCKET/"; then
    echo "$(date) Backup SUCCESS: $FILE" >> "$LOG"
    rm -f "$FILE"
else
    echo "$(date) Backup FAILED: $FILE" >> "$LOG"
fi
EOF

sudo chmod +x /usr/local/bin/amazonkart-backup.sh
```

22.3 Run a Manual Backup Test

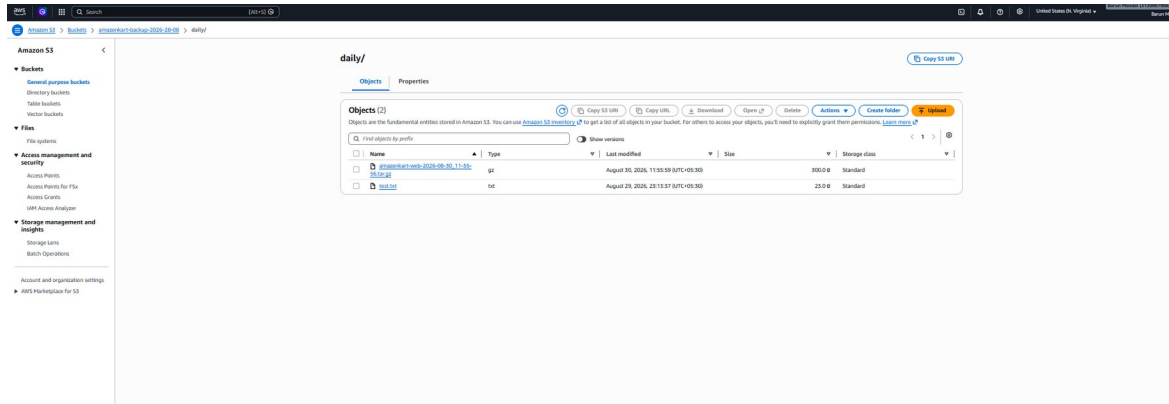
```
sudo /usr/local/bin/amazonkart-backup.sh
sudo tail -n 20 /var/log/amazonkart-backup.log
aws s3 ls s3://amazonkart-backup-2026-28-08/daily/
```

Verified success entry:

```
Sun Aug 30 11:55:58 IST 2026 Backup SUCCESS: /tmp/amazonkart-web-2026-08-30_11-55-56.tar.gz
```

Verified S3 object:

```
2026-08-30 11:55:59      300 amazonkart-web-2026-08-30_11-55-56.tar.gz
```



Evidence: daily backup archive present in S3.

22.4 Install Cron and Schedule Midnight Backup

On Amazon Linux 2023, crontab was not initially installed, so cronie was installed.

```
sudo dnf install -y cronie
sudo systemctl enable --now crond
sudo systemctl is-active crond
sudo crontab -e
```

Root crontab entry:

```
0 0 * * * /usr/local/bin/amazonkart-backup.sh
```

22.5 Verify Cron

```
sudo crontab -l
sudo systemctl status crond
```

23. Create the Golden AMI for Disaster Recovery

23.1 Console Steps

52. EC2 -> Instances -> select AmazonKart-Prod-Web01.
53. Actions -> Image and templates -> Create image.
54. AMI name: AmazonKart-Prod-Golden-AMI.
55. Create the image and wait until AMI state becomes Available.

AMI Item	Value
AMI ID	ami-05f02a718c6d21b63
Name	AmazonKart-Prod-Golden-AMI
Root snapshot	snap-086a6ce085619ab18
Root EBS	8 GB gp3
Visibility	Private
Status	Available

24. Functional Disaster Recovery Test

24.1 Launch the DR Instance

56. EC2 -> AMIs -> select ami-05f02a718c6d21b63 -> Launch instance from AMI.

57. Name: AmazonKart-DR-Recovery-Test.

58. Instance type: t3.micro.

59. Key pair: AmazonKart-Prod-Key.

60. VPC: AmazonKart-VPC.

61. Subnet: AmazonKart-Public-Subnet-1.

62. Enable public IPv4.

63. Security Group: AmazonKart-Web-SG.

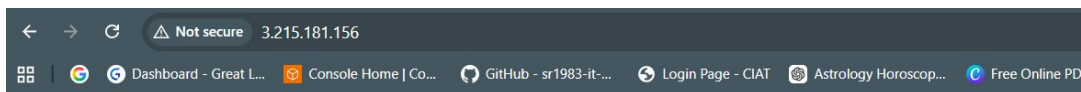
64. IAM Role: AmazonKart-EC2-Production-Role.

One tested DR instance used public IP 3.215.181.156 and private IP 10.0.1.165.

24.2 Verify the New Server

```
sudo systemctl is-active httpd
curl -I http://localhost
curl -I http://localhost/static/profile.jpeg
ls -lh /var/www/html/
ls -lh /var/www/html/static/
```

Expected: httpd active and both HTTP checks return HTTP/1.1 200 OK.



Welcome to AmazonKart

Black Friday Production Website

Server: AmazonKart-Prod-Web01

Static Content from Private S3



Evidence: DR replacement instance successfully served the AmazonKart website.

25. Test S3 Backup Restore on the DR Instance

25.1 List Available Backups

```
aws s3 ls s3://amazonkart-backup-2026-28-08/daily/
```

25.2 Initial Download Attempt and Finding

```
aws s3 cp s3://amazonkart-backup-2026-28-08/daily/amazonkart-web-2026-08-30_11-55-56.tar.gz /tmp/
```

The first attempt returned a 403 HeadObject/Forbidden error. This revealed that the production role had backup upload permission but not backup object read permission.

25.3 Corrective IAM Change

Added s3:GetObject to the backup object resource:

```
"Action": ["s3:PutObject", "s3:GetObject"],  
"Resource": "arn:aws:s3:::amazonkart-backup-2026-28-08/daily/*"
```

25.4 Retry the Download

```
aws s3 cp s3://amazonkart-backup-2026-28-08/daily/amazonkart-web-2026-08-30_11-55-56.tar.gz /tmp/
```

The retry succeeded.

25.5 Preserve Current Website and Restore Backup

```
sudo cp -r /var/www/html /var/www/html-before-restore  
  
sudo tar -xzf /tmp/amazonkart-web-2026-08-30_11-55-56.tar.gz -C /var/www  
  
ls -lh /var/www/html  
curl -I http://localhost  
curl -I http://localhost/static/profile.jpeg
```

Post-restore website returned HTTP 200, confirming that the S3 backup was usable.

26. Timed DR RTO Test

A separate clean timed launch was performed from the Golden AMI. The stopwatch started when Launch instance was clicked and stopped when the recovered website and image were available in the browser.

RTO Item	Result
Target	< 15 minutes
Recorded elapsed time	03:09:12
Project interpretation	Approximately 3 minutes 9 seconds
Result	PASS

This is the project's measured recovery evidence. Actual future recovery time may vary with AWS capacity, operator execution and data size.

27. End-to-End Validation Commands

The following command set is a useful final validation run on the production or recovered host.

```
# OS / identity  
whoami
```

```

hostname
cat /etc/os-release

# Storage
df -h

# Apache
sudo systemctl is-enabled httpd
sudo systemctl is-active httpd
curl -I http://localhost
curl -I http://localhost/static/profile.jpeg

# SSH hardening
sudo sshd -T | grep -E 'passwordauthentication|pubkeyauthentication'

# S3 access
aws s3 ls s3://amazonkart-static-2026-28-08
aws s3 ls s3://amazonkart-backup-2026-28-08/daily/

# CloudWatch Agent
sudo /opt/aws/amazon-cloudwatch-agent/bin/amazon-cloudwatch-agent-ctl -a status

# Logs
sudo ls -lh /var/log/httpd/
sudo ls -lh /var/log/secure /var/log/messages
sudo tail -n 20 /var/log/messages

# Backup
sudo systemctl is-active crond
sudo crontab -l
sudo tail -n 20 /var/log/amazonkart-backup.log

```

28. Troubleshooting We Encountered

28.1 Outbound HTTPS / AWS API Connectivity

Because the Public NACL was stateless and initially too restrictive, HTTPS return traffic could fail. Allowing ephemeral TCP 1024-65535 resolved return traffic.

```
curl -I https://aws.amazon.com
```

After correction, the request returned HTTP/2 200 and AWS API calls worked.

28.2 Static S3 Access

The bucket remained private. Instead of exposing the bucket publicly, the EC2 role used s3:GetObject and the image was copied to Apache's local static directory.

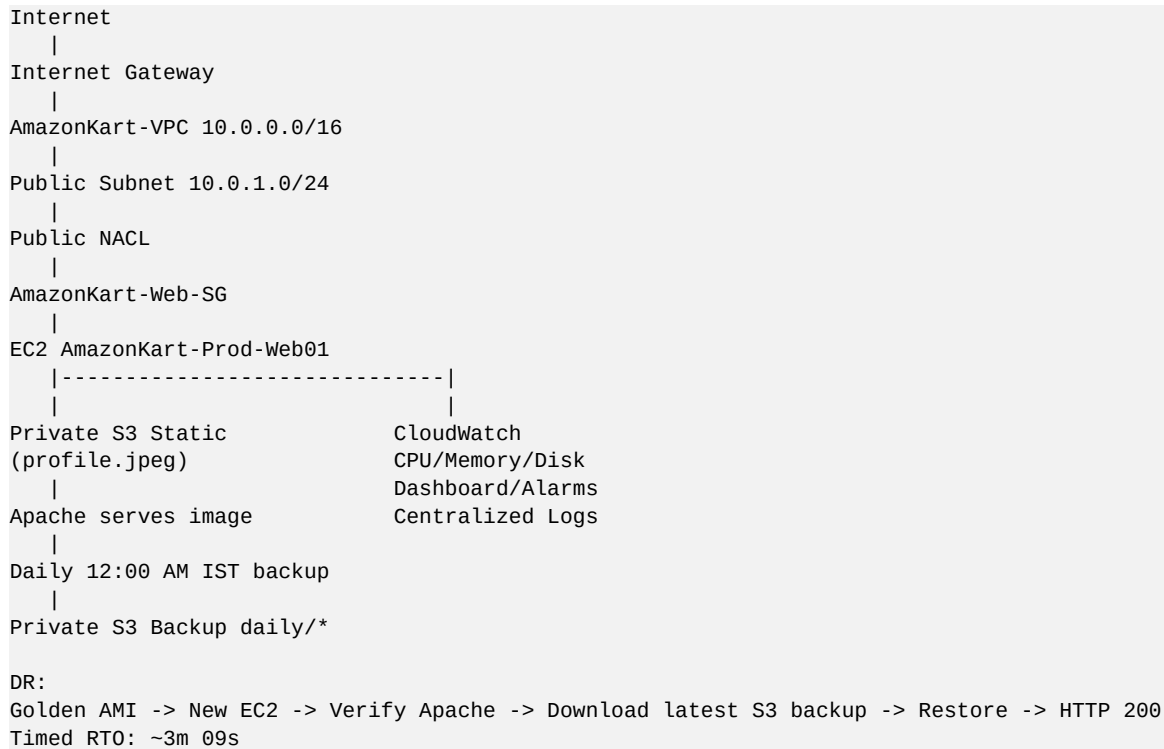
28.3 Amazon Linux 2023 Log Files

The expected challenge log names did not directly match AL2023. Apache uses access_log/error_log, and authentication events are in /var/log/secure. rsyslog was installed to create traditional system log files.

28.4 Backup Restore 403

The first restore failed because the EC2 role had PutObject but not GetObject on backup objects. Adding GetObject only to the daily/* prefix fixed recovery without granting broad S3 permissions.

29. Final Architecture Flow



30. Final Project Completion Checklist

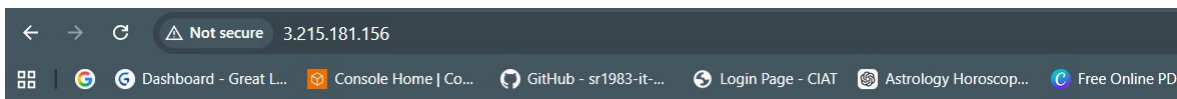
Task	Status
Custom VPC	DONE
Public subnet	DONE
Private subnet	DONE
Internet Gateway	DONE
Public route table	DONE
Public NACL	DONE
Private NACL + correct association	DONE
Security Group 22/80/443	DONE
IAM groups/users/least privilege	DONE
MFA for four IAM users	DONE
Production EC2	DONE
SSH key-only hardening	DONE
Apache production website	DONE
Private S3 static content	DONE
S3 versioning/SSE/BPA on static bucket	DONE
CloudWatch CPU/memory/disk	DONE
CloudWatch alarms	DONE
CloudWatch dashboard	DONE
Centralized access/error/auth/system logs	DONE
30-day log retention	DONE
Daily midnight S3 backup	DONE
Golden AMI	DONE
S3 restore test	DONE

31. Important Limitations to State in Interview / Submission

- No ALB or Auto Scaling was used because the challenge prohibited them. Therefore the current single EC2 web server is a single point of failure during normal operation.
- A t3.micro should not be claimed as sufficient for 100,000+ concurrent users. The design demonstrates the requested services and recovery controls under the challenge constraints.
- Port 443 is allowed, but a publicly trusted TLS certificate / HTTPS termination was not verified. Do not claim HTTPS is fully configured.
- No Elastic IP or Route 53 was used. The EC2 public IP can change after stop/start.
- SNS was deliberately not configured because the strict allowed-services list excluded it, despite a contradictory notification requirement.
- The private subnet is currently isolated and has no production workload.
- The RTO result is a measured test result, not a guaranteed SLA.

32. Evidence Appendix

Selected screenshots captured during the implementation are included below.



Welcome to AmazonKart

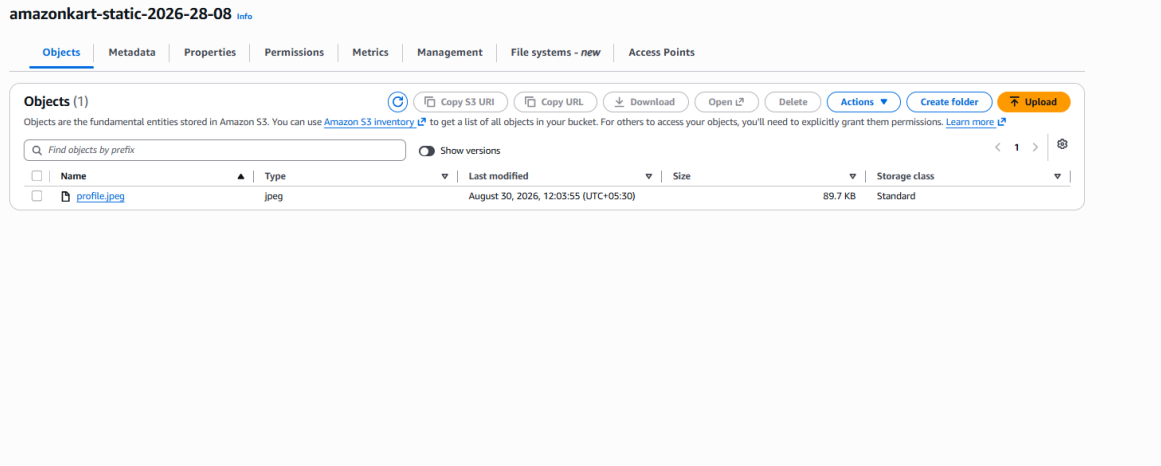
Black Friday Production Website

Server: AmazonKart-Prod-Web01

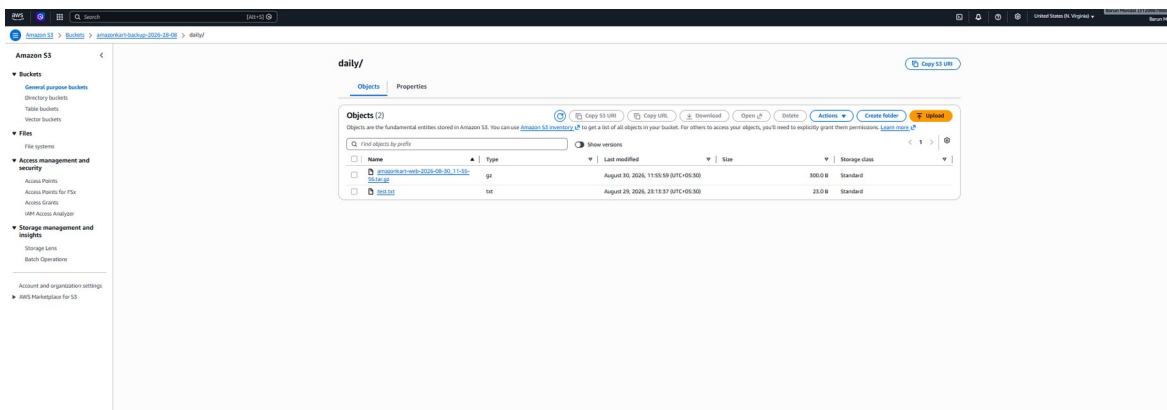
Static Content from Private S3



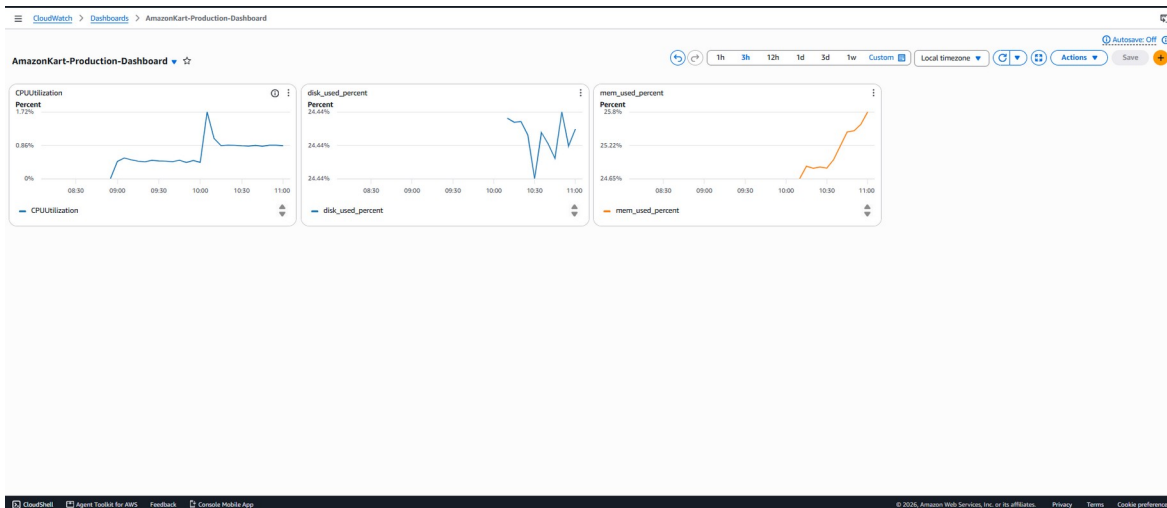
DR website recovery evidence.



Static S3 object evidence.



S3 backup evidence.



CloudWatch dashboard evidence.

Log events

You can use the filter bar below to search for and match terms, phrases, or values in your log events. [Learn more about filter patterns.](#)

Q Filter events - press enter to search

Timestamp	Message
No more records within selected time range Retry	
2026-08-30T11:34:25.433+05:30	:11 - - [29/Aug/2026:17:50:17 +0000] "HEAD / HTTP/1.1" 200 - "-" "curl/7.17.0"
2026-08-30T11:34:25.433+05:30	106.219.255.188 - - [29/Aug/2026:17:57:14 +0000] "GET / HTTP/1.1" 200 227 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/152.0.8.0 Safari/537.36"
2026-08-30T11:34:25.433+05:30	106.219.255.188 - - [29/Aug/2026:17:57:18 +0000] "GET /favicon.ico HTTP/1.1" 404 236 "http://18.204.23.58/" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/152.0.8.0 Safari/537.36"
2026-08-30T11:34:25.434+05:30	106.219.255.188 - - [29/Aug/2026:17:58:06 +0000] "-" 408 - "-" "-"
2026-08-30T11:34:25.434+05:30	106.219.255.188 - - [29/Aug/2026:17:58:06 +0000] "-" 408 - "-" "-"
2026-08-30T11:34:25.434+05:30	3.71.7.238 - - [29/Aug/2026:18:27:36 +0000] "GET / HTTP/1.1" 200 227 "-" "Mozilla/5.0 (Linux; Android 4.3; SPH-L710 Build/255152) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/52.0.1700.99 Mobile Safari/537.36"
2026-08-30T11:34:25.434+05:30	106.219.255.168 - - [29/Aug/2026:19:29:25 +0000] "GET / HTTP/1.1" 304 - "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/152.0.8.0 Safari/537.36"
2026-08-30T11:34:25.434+05:30	106.219.255.168 - - [29/Aug/2026:19:29:26 +0000] "GET / HTTP/1.1" 304 - "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/152.0.8.0 Safari/537.36"
2026-08-30T11:34:25.434+05:30	106.219.255.168 - - [29/Aug/2026:19:29:27 +0000] "GET / HTTP/1.1" 304 - "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/152.0.8.0 Safari/537.36"
2026-08-30T11:34:25.434+05:30	106.219.255.168 - - [29/Aug/2026:19:30:14 +0000] "-" 408 - "-" "-"
2026-08-30T11:34:25.434+05:30	146.190.169.94 - - [30/Aug/2026:03:36:50 +0000] "GET / HTTP/1.1" 200 227 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/128.0.8.0 Safari/537.36"
2026-08-30T11:34:25.434+05:30	172.236.228.229 - - [30/Aug/2026:03:46:46 +0000] "\x16\x03\x03\x03\x03" 400 266 "-" "-"
2026-08-30T11:34:25.434+05:30	172.236.228.229 - - [30/Aug/2026:03:46:47 +0000] "\x16\x03\x03\x03\x03" 400 266 "-" "-"
2026-08-30T11:34:25.434+05:30	45.79.182.179 - - [30/Aug/2026:04:29:15 +0000] "GET / HTTP/1.1" 200 227 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 11_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/108.0.8.0 Safari/537.36"
2026-08-30T11:34:25.434+05:30	45.79.287.71 - - [30/Aug/2026:04:40:15 +0000] "GET / HTTP/1.1" 200 227 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 11_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/108.0.8.0 Safari/537.36"
2026-08-30T11:34:25.434+05:30	66.228.53.157 - - [30/Aug/2026:04:40:47 +0000] "GET / HTTP/1.1" 200 227 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 11_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/108.0.8.0 Safari/537.36"
2026-08-30T11:34:25.434+05:30	198.235.24.11 - - [30/Aug/2026:04:47:02 +0000] "\x16\x03\x03" 400 266 "-" "-"
2026-08-30T11:34:25.434+05:30	198.235.24.11 - - [30/Aug/2026:04:47:03 +0000] "\x16\x03\x03" 400 266 "-" "-"
2026-08-30T11:34:25.434+05:30	20.121.60.113 - - [30/Aug/2026:04:48:13 +0000] "GET /developmentserver/ectadatauploader HTTP/1.1" 404 236 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 11_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/128.0.8.0 Safari/537.36"
2026-08-30T11:34:25.434+05:30	64.89.163.130 - - [30/Aug/2026:05:13:12 +0000] "GET / HTTP/1.1" 200 227 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 10_15_6) AppleWebKit/605.1.15 (KHTML, like Gecko) Version/13.6.6 Safari/605.1.15"
2026-08-30T11:34:25.434+05:30	64.89.163.130 - - [30/Aug/2026:05:15:52 +0000] "GET /.css HTTP/1.1" 404 236 "-" "Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/58.0.3029.110 Safari/537.36"
2026-08-30T11:34:30.640+05:30	172.104.11.4 - - [30/Aug/2026:05:44:54 +0000] "GET / HTTP/1.1" 200 227 "-" "Mozilla/5.0 (Macintosh; Intel Mac OS X 11_3) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/108.0.8.0 Safari/537.36"
No more records within selected time range Auto retry paused. Resume	

CloudWatch log evidence.

Welcome to AmazonKart

Black Friday Production Website

Server: AmazonKart-Prod-Web01

Static Content from Private S3



Production website with private-S3 content.